VM-660-41

HIGH-SPEED MODULAR GEARBOX

Eccentric-Cycloidal Gearing · T = 100 kNm · U = 50 · Ø 660 mm · 7,000 rpm

Sanctions-Free Supply · No Single-Source Dependency · Full Engineering Customisation

100	**7,000**	**93–95**	**490**	**10,000**
kNm	rpm	%	kg	h
Nominal Torque	*Max Input Speed*	*Efficiency*	*Weight*	*Service Life*

THE SUPPLY CHAIN IMPERATIVE

Why global OEMs and EPC contractors are re-evaluating gearbox sourcing

European suppliers restricted

Flender, SEW-Eurodrive, Bosch Rexroth — under export-control regimes for large parts of the global market. Parallel-channel lead times 6–18 months, no warranty, no after-sales support.

Single-source risk

Over 70% of heavy gearbox supply in the 100 kNm class comes from three European manufacturers. One geopolitical event can halt entire production lines. Multi-sourcing is now a procurement mandate.

Nearshoring trend

EPC contractors across the Middle East, Asia, Africa and Latin America actively seek non-European, non-Chinese alternatives with certified quality and full documentation.

ExCentro VM-660-41 — certified, fully documented, sanctions-free alternative to European premium gearboxes

No sanctions exposure

Manufactured outside restricted supply chains. No export licence required for any market.

Drop-in replacement

Adaptable to Flender / Bosch Rexroth / SEW interface standards. Full engineering package.

4–8 week delivery

vs 6–18 months for European equivalents via parallel import routes.

Performance on par

93–95% efficiency, 7,000 rpm input — meets or exceeds Flender NGW / SEW P-series.

ECCENTRIC-CYCLOIDAL GEARING

What makes this design fundamentally different

EC
gearing

7,000 rpm input — unique in the 100 kNm class

All major competitors (Flender, SEW, Bonfiglioli, Brevini) are limited to 1,500–4,500 rpm at this torque. Enables pairing with compact high-speed VFD motors — cutting drivetrain CAPEX by 20–25%.

Specific mass 0.0035 kg/Nm — 3–6× lighter than conventional

490 kg at 100 kNm. Conventional helical gearboxes at the same torque: 1,500–3,000 kg. Critical for cranes, marine, offshore and any weight-constrained application.

Multi-pair tooth contact — extended service life

Load is shared across multiple tooth pairs simultaneously; specific pressure is 3–5× lower than involute gearing. Result: 10,000 h service life vs the 8,000 h typical of European equivalents.

TECHNICAL SPECIFICATIONS

VM-660-41 · High-Speed High-Torque Modular Gearbox

Value	Unit	Parameter
100	kNm	Nominal Output Torque
140	kNm	Max Output Torque
50		Gear Ratio U
7,000	rpm	Max Input Speed
490	kg	Weight
10,000	h	Service Life
660	mm	Outer Diameter
330	mm	Height
93–95	%	Efficiency
0.0035	kg/Nm	Specific Mass

COMPETITIVE COMPARISON

Heavy gearboxes ~100 kNm class — ExCentro vs world leaders

Parameter	ExCentro VM-660-41	Flender / SEW (Europe)	Guomao / NGC (China)	Conv. Helical (legacy)
Gearing type	**EC (eccentric-cycloidal)**	Planetary / bevel-helical	Planetary / cylindrical	Helical (involute)
Weight, kg	**490**	500–600	600–700	1,500–3,000
Max input speed	**7,000 rpm ★**	3,000–4,500 rpm	up to 3,000 rpm	up to 1,500 rpm
Efficiency	**93–95%**	94–96%	90–93%	96–97%
Availability (2026)	**Sanctions-free ✓**	Restricted / 6–18 mo	3–4 mo, CNY risk	Available, obsolete
Customisation	**High — ExCentro platform**	Catalogue only	Limited	Partial
Delivery	**4–8 weeks**	6–18 months	3–4 months	In stock

★ 7,000 rpm at 100 kNm — unmatched across all major OEM catalogues (Flender, SEW, Bosch Rexroth, Bonfiglioli, Brevini)

OIL & GAS / CHEMICAL INDUSTRY

Top Drive Systems · Heavy Pump Drives · Winches · Extruders

Typical Applications

- Top Drive Systems (TDS) — replacement for Bosch Rexroth GFT series
- Heavy crude pump drives — refineries and pipeline stations
- Offshore winches and mooring systems
- Extruder and mixer drives — petrochemicals, polymers
- Pipeline valve actuators DN>600

Key clients: ADNOC · Saudi Aramco · PETRONAS · KOC · EPC contractors (Wood, TechnipFMC, Worley)

−30–40% drivetrain weight

490 kg vs 1,500–2,000 kg conventional. Lower structural load on the rig — reduced substructure CAPEX.

Smaller, cheaper VFD motor

7,000 rpm input → high-speed motor instead of low-speed — 20–25% motor CAPEX saving per drivetrain.

10,000 h life — fewer shutdowns

Flender NGW benchmark: 8,000 h. Each avoided shutdown = +2–5% uptime. Critical for 24/7 offshore operations.

Sanctions-free sourcing

No export licence required. Deliverable to the Middle East, Asia, Africa. Replaces Flender/Rexroth on a 4–8 week lead time.

MARINE, PORTS & HEAVY LIFTING

Ship Winches · Portal Cranes · Thrusters · Naval Drives

Weight is cost — 490 kg saves 1–1.5 t per unit

On a vessel, every kilogram saved in machinery converts directly into payload capacity or fuel efficiency. 490 kg vs 1,500–2,000 kg of a conventional helical gearbox at the same torque.

Non-European, non-Chinese sourcing

Naval and offshore projects in the Middle East, South-East Asia and Africa require independence from both European (restricted) and Chinese (geopolitically sensitive) sources. ExCentro fills this gap.

High input speed → compact motor room

7,000 rpm enables a high-speed motor, shrinking the motor-room footprint — critical for tight hull sections in naval and offshore designs.

4+ years between overhauls

10,000 h service life. For a port crane on two shifts: over 4 years without a gearbox overhaul. Brevini/Dana standard: 6,000–8,000 h.

Vs Brevini / Dana (Italy/USA) — parallel import 6–9 months, +35–50% price | Vs Guomao (China) — available, but limited customisation, +200–400 kg weight

ENERGY & MACHINE TOOLS

Coal Mill Drives · Power Plant Auxiliaries · High-Speed Machining

Power Generation

Thermal plants · Hydro · Water infrastructure

Coal mill drives — power plants

93–95% efficiency in continuous duty. Sanctions-free alternative to Flender KMPS. Delivery 4–8 weeks vs 6–18 months parallel import.

Nuclear-grade auxiliary equipment

10,000 h life, full documentation, localisable manufacturing — meets nuclear-industry localisation and traceability requirements.

Pump station infrastructure

490 kg compactness reduces structural load on pump-house floors. Effect: −15–20% civil-infrastructure CAPEX.

High import-substitution demand · Strict reliability and documentation requirements

Machine Tools & OEM

Heavy machining centres · Defence manufacturing · Robotics

Spindle / main drive

7,000 rpm direct coupling to a high-speed VFD motor — no intermediate stage. Vs Bosch Rexroth: equivalent layout without supply restrictions.

Heavy machining centres

Specific mass 0.0035 kg/Nm = minimum rotational inertia → fast acceleration/deceleration. Cycle-time reduction of 8–12%.

Defence manufacturing & robotics

Ø660×330 mm at 100 kNm — a unique compactness ratio for heavy manipulators and special-purpose machines.

Key argument: 7,000 rpm at 100 kNm — absent from all Flender, SEW, Bosch Rexroth catalogues

ADAPTATION & OEM PARTNERSHIP

The Module can be adapted to different interface dimensions, gear ratios and torque ranges

Gear Ratio

U = 20 to 80+ in a single stage

Interface & Mounting

Flanges, shafts and centering adapted to customer or OEM standards

Torque Range

Scalable series — 50 to 200+ kNm

Protection Class

IP65+ · Marine · ATEX hazardous area

OEM Benefits — Total Cost of Ownership

−30–40% system weight

490 kg vs 1,500–3,000 kg conventional → lighter frames, foundations and installation equipment

High-speed motor saving

7,000 rpm: a smaller motor at −20–25% cost → lower total drivetrain CAPEX

Lower OPEX

10,000 h life → fewer planned shutdowns, smaller spare-parts inventory

Sanctions-free supply

No export licence, no geopolitical risk — deliverable to any market worldwide

ExCentro

VM-660-41 · Eccentric-Cycloidal Gearbox · Sanctions-Free Supply

7,000 rpm | 100 kNm | 490 kg | 10,000 h | No sanctions risk

Ready to qualify ExCentro as your alternative gearbox supplier?

Request Technical Package & Commercial Proposal

Technical Inquiry

Questionnaire → sizing

NDA & Negotiations

Chat with ExCentro Assistant

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